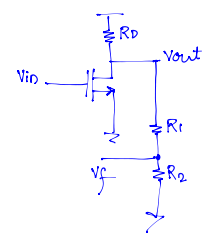
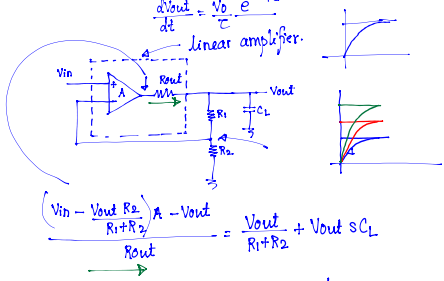


Slew rate :-

$$\begin{aligned}
 \frac{V_{out}(s)}{V_{in}} &= \frac{1/s}{1/s + R} \\
 &= \frac{1}{1 + sRC} = \frac{1}{1 + s\tau} \\
 V_{out}(s) &= \frac{1}{1 + s\tau} \cdot V_{in}(s) \\
 &= \frac{1}{1 + s\tau} \cdot \frac{V_0}{s} \\
 &= V_0 \left(\frac{1}{s} + \frac{\tau}{1 + s\tau} \right) \\
 &= V_0 \left(\frac{1}{s} + \frac{\tau}{1 + s\tau} \right) \\
 V_{out}(t) &= V_0 \left(u(t) - e^{-t/\tau} u(t) \right) \\
 &= V_0 \left(1 - e^{-t/\tau} \right) u(t) \\
 \frac{dV_{out}}{dt} &= \frac{V_0}{\tau} e^{-t/\tau}
 \end{aligned}$$

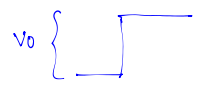
$$\begin{aligned}
 A(1 + s\tau) + Bs &= 1 \\
 A &= 1 \\
 s &= -\frac{1}{\tau} \\
 B &= \frac{1}{\tau} \\
 \beta &= -\tau \\
 e^{at} &= \frac{1}{s-a}
 \end{aligned}$$



$$\begin{aligned}
 A V_{in} - \frac{V_{out} R_2}{R_1 + R_2} &= \frac{V_{out} R_{out}}{R_1 + R_2} + V_{out} s C_L R_{out} \\
 A V_{in} &= \left[\frac{A R_2}{R_1 + R_2} + 1 + \frac{R_{out}}{R_1 + R_2} + s C_L R_{out} \right] V_{out}
 \end{aligned}$$

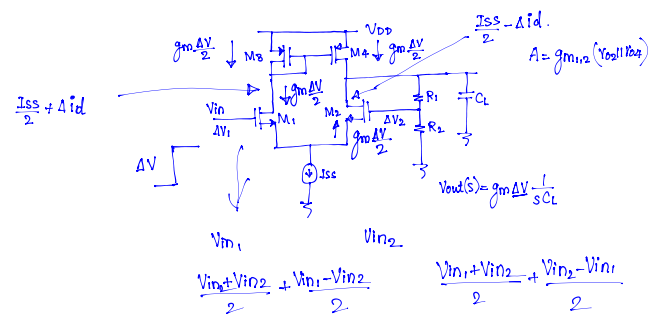
$$\frac{V_{out}}{V_{in}} = \frac{A}{1 + \frac{A R_2}{R_1 + R_2} + s C_L R_{out}}$$

$$\frac{1}{1 + j\omega \frac{1}{\omega_0 (1 + \beta A)}}$$

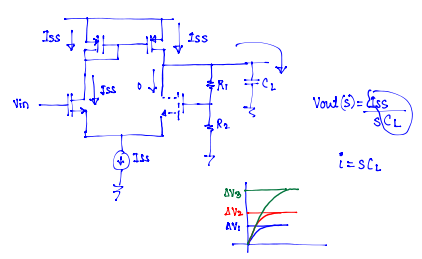


$$V_{out}(t) = V_0 \frac{A}{1 + A \frac{R_2}{R_1 + R_2}} \left[1 - e^{-t/\tau} \right]$$

$$\tau = C_L \frac{R_{out}}{1 + A \frac{R_2}{R_1 + R_2}}$$



$$\begin{aligned}
 V_{in1} &= \frac{V_{in1} + V_{in2}}{2} + \frac{V_{in1} - V_{in2}}{2} \\
 V_{in2} &= \frac{V_{in1} + V_{in2}}{2} - \frac{V_{in1} - V_{in2}}{2}
 \end{aligned}$$



$$\begin{aligned}
 i &= C \frac{dv}{dt} \\
 \frac{1}{C} \int i dt &= dv \\
 \frac{dv}{dt} &= \frac{1}{C} i \\
 \left. \frac{dv}{dt} \right|_{max} &= \frac{I_{SS}}{C_L} = 5 \text{ V}/\mu\text{s}
 \end{aligned}$$